MATH 180 Strategies of Problem Solving

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Outline

1. SoPS Lecture 17
   - Objects and statements
   - Describing sets
   - Subsets and the empty set
   - Sets containing sets
2. Set algebra
   - Union, intersection, and difference
   - Counting and power sets
   - Ordered pairs and products
   - Quantifiers and proofs
3. Why is it called naive?
   - Russell's paradox

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What are we learning to do?

Our questions

- What objects does an expression describe?
- Does a sentence make mathematical sense?
- If it makes sense, is it true?
- How can we justify our answer?

A false statement can still make perfectly good mathematical sense.

A first sorting puzzle

You know ordinary arithmetic. Classify each line.

$$2+3=5 \qquad 2+3=8 \qquad 2+3 \qquad 2+ \ =5$$

Four possible responses

- A meaningful true statement.
- A meaningful false statement.
- A meaningful expression, but no claim to evaluate.
- An incomplete or undefined expression.

A first sorting puzzle — discussion

- $2 + 3 = 5$ is a true statement.
- $2 + 3 = 8$ is a false statement.
- $2 + 3$ names a number. It makes sense, but is not a statement.
- $2 + \ = 5$ is incomplete as written. We could repair it by introducing a variable.

One more possibility

$x + 3 = 5$ is an **open sentence**. Its truth depends on x.

Sets

Working description

A **set** is a collection of distinct objects, called its **elements** or **members**.

- $A = \{1, 2, 3\}$ is a set of numbers.
- $V = \{a, e, i, o, u\}$ is a set of letters.
- A set can contain game positions, people, or other sets.

We must be able to determine what counts as an element.

Membership

Let $A = \{1, 2, 3\}$.

$2 \in A$ means that 2 is an element of A.

$5 \notin A$ means that 5 is not an element of A.

Reading aloud

Read $\in$ as "is an element of." This helps distinguish membership from other relationships.

Which lines make sense?

Let $A = \{1, 2, 3\}$. We use numbers as ordinary numbers.

Discuss with a partner

- $2 \in A$
- $5 \in A$
- $A + 2 = 5$
- A

Which lines make sense? — discussion

Let $A = \{1, 2, 3\}$. We use numbers as ordinary numbers.

- $2 \in A$: meaningful and true.
- $5 \in A$: meaningful and false.
- $A + 2 = 5$: we have not defined addition of a set and a number.
- A: a meaningful name for a set, but no statement.

Conventions matter

Our convention

Numbers and letters are ordinary objects. We do not use a special set-theoretic encoding of numbers.

- Expressions like $2 \subseteq A$ will be undefined in our language, because 2 is not being treated as a set.
- Set membership $x \in A$ requires A to be a set. The object x may itself be a set.
- An unfamiliar but well-defined expression may be false. Unfamiliarity alone does not make it meaningless.

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Order and repetition

$$\{1, 2, 3\} = \{3, 1, 2\} = \{1, 1, 2, 3, 3\}.$$

Equality of sets

Two sets are equal exactly when they have the same elements.

- Order does not matter.
- Listing an element again does not create another element.

A set of dice outcomes and a sequence of dice rolls record different information.

The same set?

For each proposed equality, decide whether it is true.

Discuss with a partner

- $\{2, 4, 6\} = \{6, 2, 4\}$
- $\{1, 1, 2\} = \{1, 2, 2\}$
- $\{1, 2\} = \{3\}$
- $\{1, 2\} = 3$

The same set? — discussion

For each proposed equality, decide whether it is true.

- True: the elements are identical.
- True: each side has exactly the elements 1 and 2.
- False: 1 belongs to the left side and not the right.
- False under our convention: a two-element set is not the number 3. Equality can compare objects of different kinds.

Set-builder notation

$$E = \{x \in \mathbb{Z} : x \text{ is even}\}.$$

- $x \in \mathbb{Z}$ specifies the allowed objects.
- The colon means "such that.
- The condition selects which allowed objects belong.

Finite example

$\{x \in \mathbb{Z} : 0 < x < 5\} = \{1, 2, 3, 4\}$.

Some familiar infinite sets

$$\mathbb{N} = \{1, 2, 3, \ldots\},$$
$$\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\},$$
$$\mathbb{Q} = \{a/b : a, b \in \mathbb{Z},\ b \neq 0\},$$
$$\mathbb{R} = \text{the set of real numbers.}$$

In these slides, $0 \notin \mathbb{N}$. Some books use a different convention.

Infinite sets need a description that specifies every element.

Reading a set description

Let $B = \{x \in \mathbb{Z} : x^2 < 10\}$.

Discuss with a partner

- List all elements of B.
- Is $\sqrt{2} \in B$?
- Is $-3 \in B$?
- Is $\{x \in \mathbb{Z} : x^2\}$ a complete set description?

Reading a set description — discussion

Let $B = \{x \in \mathbb{Z} : x^2 < 10\}$.

- $B = \{-3, -2, -1, 0, 1, 2, 3\}$.
- No. $\sqrt{2}$ fails the requirement $x \in \mathbb{Z}$.
- Yes, because $(-3)^2 = 9 < 10$.
- No. x^2 is a number, not a condition to test.

Conditions versus output expressions

$$\{x \in \mathbb{Z} : x^2 < 10\} \qquad \{x^2 : x \in \mathbb{Z}\}$$

- The first selects integers satisfying a condition.
- The second collects the outputs x^2 for all integers x.
- The second set is $\{0, 1, 4, 9, 16, \ldots\}$.

Check

Why does the second set contain 9 only once, even though $3^2 = (-3)^2$?

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Subsets

Definition

$A \subseteq B$ means that **every element of A is also an element of B.**

$$\{1, 3\} \subseteq \{1, 2, 3\}.$$

- $A \subseteq A$ for every set A.
- $A \subsetneq B$ means $A \subseteq B$ and $A \neq B$.
- We use $\subseteq$ to allow equality and $\subsetneq$ to exclude it.

The symbol $\subset$ has different conventions in different books.

Membership versus inclusion

Let $A = \{1, 2, 3\}$.

$$2 \in A \qquad \{2\} \subseteq A.$$

What are we comparing?

- $x \in A$ asks whether x is one of the elements of A.
- $B \subseteq A$ asks whether every element of B belongs to A.

$\{2\}$ is a set with one element. Its element is 2.

Membership or subset?

Let $A = \{1, 2, 3\}$. Classify each line.

Discuss with a partner

- $1 \in A$ and $\{1\} \in A$
- $\{1\} \subseteq A$ and $A \subseteq A$
- $1 \subseteq A$
- $A \in A$

Membership or subset? — discussion

Let $A = \{1, 2, 3\}$. Classify each line.

- True and false, respectively. The elements of A are numbers.
- Both true.
- Undefined in our convention: the left side must be a set.
- Meaningful and false. None of the listed elements is A.

The empty set

Definition

The **empty set**, written $\varnothing$ or $\{\}$, has no elements.

$$\{x \in \mathbb{R} : x^2 = -1\} = \varnothing.$$

- $\varnothing$ is a set.
- $\{\varnothing\}$ has one element: the empty set.
- $\varnothing \neq \{\varnothing\}$.

Question

Is a box containing one empty box itself empty?

Why the empty set is a subset of every set

For any set A,

$$\varnothing \subseteq A.$$

Reason

To disprove this, we would need an element of $\varnothing$ that fails to belong to A. There is no element of $\varnothing$ at all.

This is an example of **vacuous truth**.

"Every dragon in this room wears a hat is true if there are no dragons in this room.

Empty-set check

Let $A = \{1, 2\}$.

Discuss with a partner

- $\varnothing \subseteq A$
- $\varnothing \in A$
- $\varnothing \in \{\varnothing\}$
- $\{\varnothing\} \subseteq A$

Empty-set check — discussion

Let $A = \{1, 2\}$.

- True: the empty set is a subset of every set.
- False: the only elements of A are 1 and 2.
- True: it is the one listed element.
- False: this would require $\varnothing \in A$.

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Sets can be elements

Let

$$C = \{1, \{2, 3\}, \varnothing\}.$$

- The elements of C are 1, $\{2, 3\}$, and $\varnothing$.
- $\{2, 3\} \in C$ is true.
- $2 \in C$ is false.
- $\{2, 3\} \subseteq C$ is false.

Braces are structure

A set containing a set does not automatically contain the elements of that inner set.

Braces change the answer

Let $D = \{1, \{1\}, \{2, 3\}\}$.

Discuss with a partner

- $1 \in D$ and $\{1\} \in D$
- $\{1\} \subseteq D$
- $\{\{1\}\} \subseteq D$
- $\{2, 3\} \subseteq D$

Braces change the answer — discussion

Let $D = \{1, \{1\}, \{2, 3\}\}$.

- Both true: both objects occur in the list.
- True: its only element, 1, belongs to D.
- True: its only element, $\{1\}$, belongs to D.
- False: neither 2 nor 3 is an element of D.

A relation between the two symbols

Always true

For every object x and every set A,

$$x \in A \iff \{x\} \subseteq A.$$

- The only element of $\{x\}$ is x.
- So checking every element of $\{x\}$ means checking x.

Careful

This does not say $x \in A$ is equivalent to $\{x\} \in A$.

A short group challenge

Construct a set A satisfying each request separately.

- $1 \in A$ but $\{1\} \notin A$.
- $\{1\} \in A$ but $1 \notin A$.
- Both $1 \in A$ and $\{1\} \in A$.
- $\varnothing \in A$ and A has exactly two elements.

Compare constructions. Can different groups give different correct answers?

A short group challenge — examples

- $A = \{1\}$.
- $A = \{\{1\}\}$.
- $A = \{1, \{1\}\}$.
- $A = \{\varnothing, 7\}$.

Another check

Why does $\{\varnothing, \varnothing\}$ fail?

It has only one element. Repetition does not count.

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Union and intersection

For sets A and B,

$$A \cup B = \{x : x \in A \text{ or } x \in B\},$$
$$A \cap B = \{x : x \in A \text{ and } x \in B\}.$$

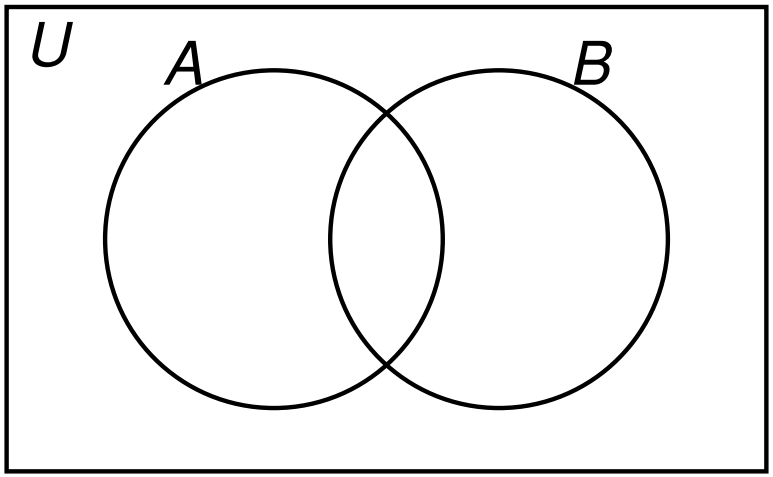
The mathematical word or

An element belongs to $A \cup B$ if it belongs to either set **or to both**.

Example: if $A = \{1, 2, 3\}$ and $B = \{3, 4\}$,

$$A \cup B = \{1, 2, 3, 4\}, \qquad A \cap B = \{3\}.$$

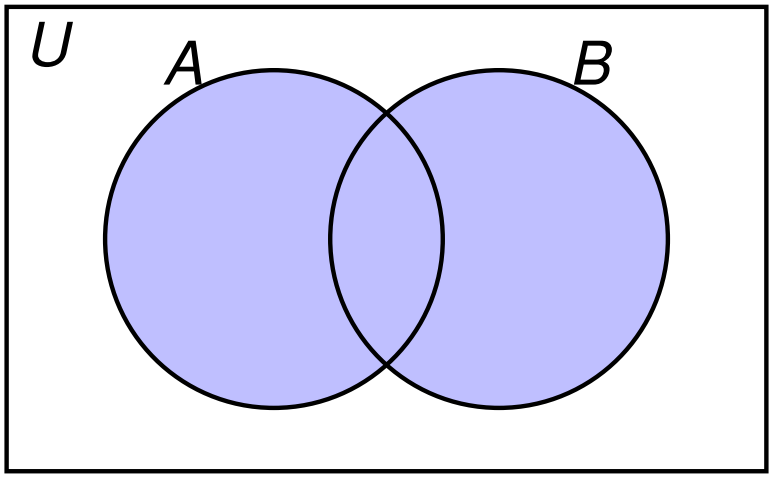
Reading a Venn diagram



The rectangle is the universe U. The circles represent sets A and B.

Each position represents a possible element. The overlap represents elements belonging to both sets.

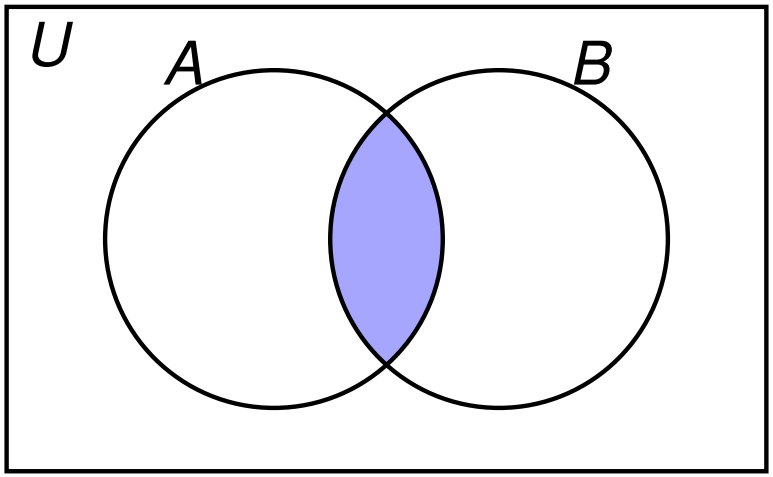
Union in a Venn diagram



$$x \in A \cup B \iff x \in A \text{ or } x \in B.$$

The shaded region includes the overlap. Mathematical “or” allows both.

Intersection in a Venn diagram



$$x \in A \cap B \iff x \in A \text{ and } x \in B.$$

Check: Does a point in A but outside B belong to the shaded set?

Difference and complement

$$A \setminus B = \{x \in A : x \notin B\}.$$

If $A = \{1, 2, 3\}$ and $B = \{3, 4\}$,

$$A \setminus B = \{1, 2\}, \qquad B \setminus A = \{4\}.$$

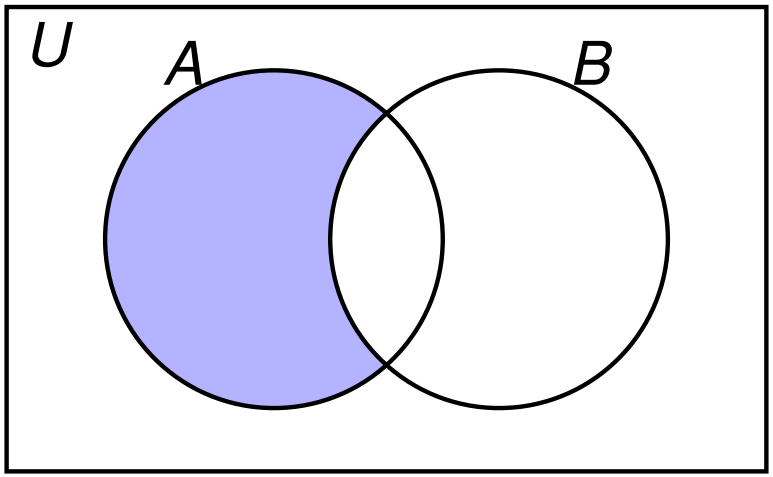
Complement needs a universe

If all objects under discussion lie in a fixed set U, then

$$A^c = U \setminus A.$$

Without a specified or understood U, the complement is ambiguous.

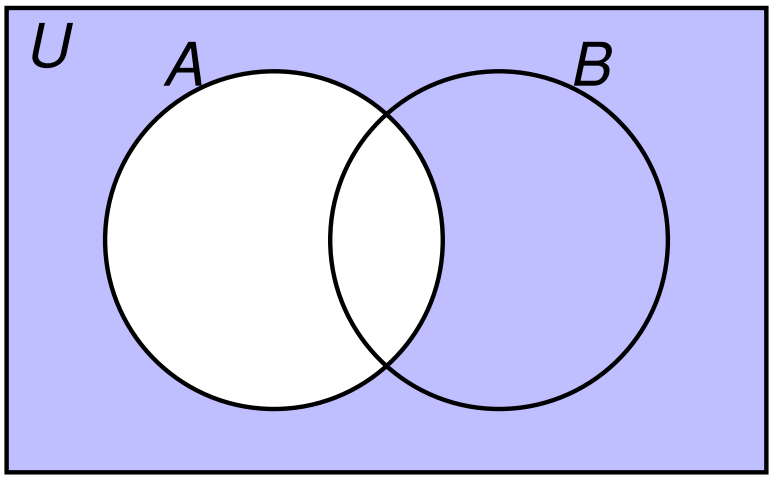
Difference in a Venn diagram



$$A \setminus B = \{x \in A : x \notin B\}.$$

Check: Where would we shade $B \setminus A$ instead?

Complement in a Venn diagram



The shaded region is $A^c = U \setminus A$.

Check: Why does part of B belong to A^c, while another part does not?

Calculating with sets

Let $U = \{1, 2, 3, 4, 5, 6\}$, $A = \{1, 2, 4\}$, $B = \{2, 3, 4\}$.

Discuss with a partner

- Find $A \cup B$ and $A \cap B$.
- Find $A \setminus B$ and $B \setminus A$.
- Find A^c.
- Is $A \cap B = 2$ true?

Calculating with sets — discussion

Let $U = \{1, 2, 3, 4, 5, 6\}$, $A = \{1, 2, 4\}$, $B = \{2, 3, 4\}$.

- $A \cup B = \{1, 2, 3, 4\}$ and $A \cap B = \{2, 4\}$.
- $A \setminus B = \{1\}$ and $B \setminus A = \{3\}$.
- $A^c = \{3, 5, 6\}$.
- False. The intersection is a set, not the number of its elements.

Which expressions mean something?

Let $A = \{1, 2\}$ and $B = \{2, 3\}$.

Discuss with a partner

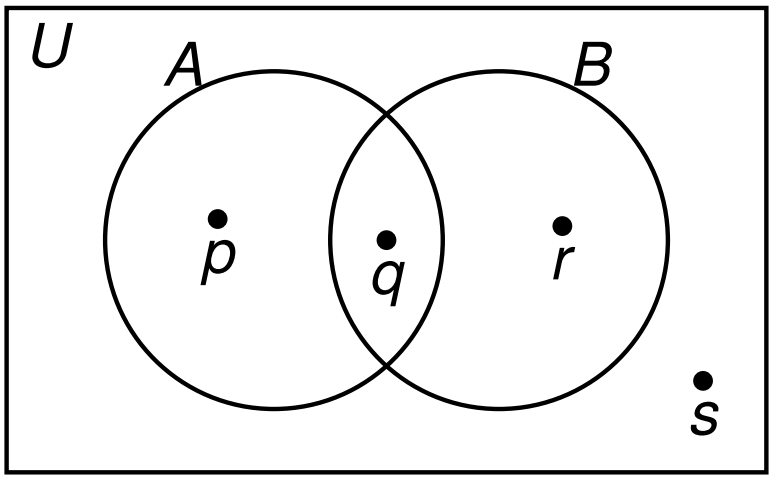
- $A \cup B$
- $A \cup 2$
- $2 \in A \cap B$
- $A \in B \cup \{A\}$

Which expressions mean something? — discussion

Let $A = \{1, 2\}$ and $B = \{2, 3\}$.

- Meaningful set expression: $\{1, 2, 3\}$. No truth value by itself.
- Undefined: union requires two sets in our convention.
- Meaningful and true, since $A \cap B = \{2\}$.
- Meaningful and true. The union has A as an element.

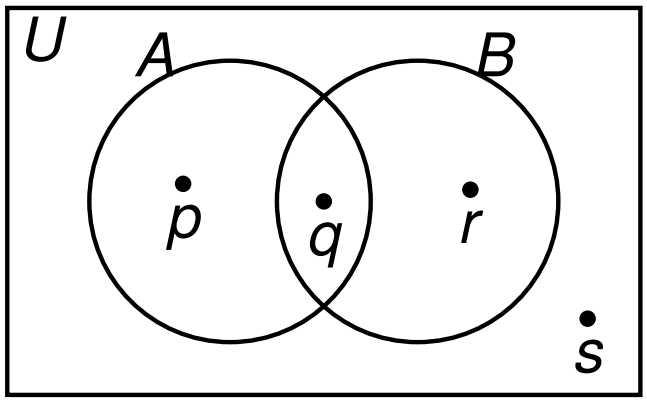
Venn diagram check



Which points belong to $A \cup B$, $A \cap B$, $A \setminus B$, and $(A \cup B)^c$?

Are $p \in A$ and $p \subseteq A$ equally meaningful? Treat p as a point, not a set.

Venn diagram check — answers



Among the labeled points: p, q, r belong to $A \cup B$, q belongs to $A \cap B$, p belongs to $A \setminus B$, and s belongs to $(A \cup B)^c$.

$p \in A$ is true. We have not defined $p \subseteq A$ for a point.

Disjoint sets

Definition

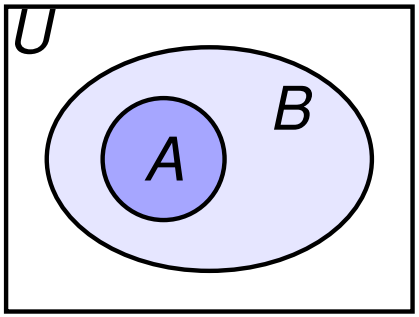
Sets A and B are **disjoint** if $A \cap B = \varnothing$.

- $\{1, 3, 5\}$ and $\{2, 4, 6\}$ are disjoint.
- $\{1, 2\}$ and $\{2, 3\}$ are different but not disjoint.
- $\{\varnothing\}$ and $\varnothing$ are disjoint.

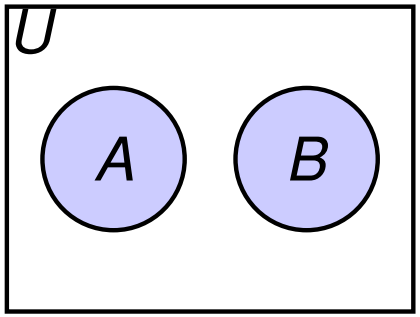
Check

Can a nonempty set be disjoint from itself? Explain.

Inclusion and disjointness in pictures



$A \subseteq B$



$A \cap B = \varnothing$

The left picture says every element of A belongs to B.

It does not assert $A \in B$. Circles contain points representing elements, not automatically the circles themselves.

Identities worth testing

For arbitrary sets A and B, which are always true?

- $A \cup B = B \cup A$.
- $A \cap B = B \cap A$.
- $A \setminus B = B \setminus A$.
- $A \cup \varnothing = A$ and $A \cap \varnothing = \varnothing$.

For a false claim, find a small counterexample.

Identities worth testing — discussion

- Union and intersection are commutative: the first two are true.
- Difference need not be commutative. Take $A = \{1\}$ and $B = \{2\}$. Then $A \setminus B = \{1\}$ and $B \setminus A = \{2\}$.
- Both empty-set identities are true.

A counterexample

One example where a universal claim fails disproves that claim. Many successful examples do not prove it.

De Morgan's laws

With a fixed universe U,

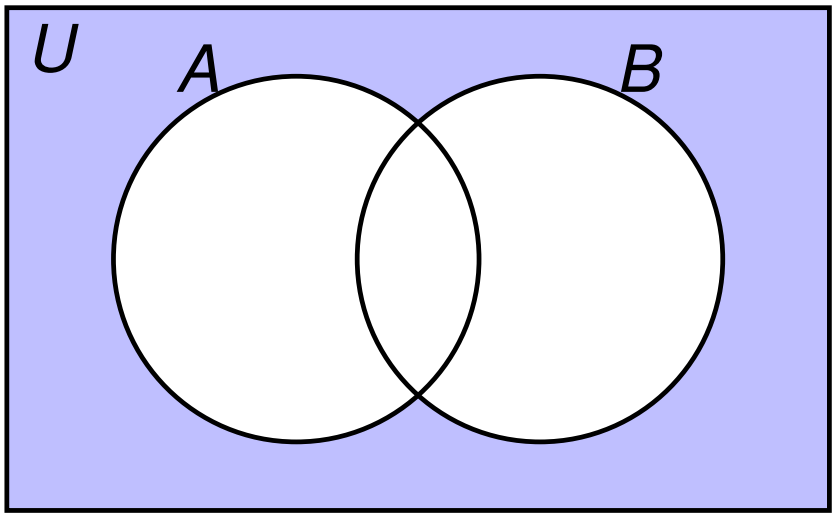
$$(A \cup B)^c = A^c \cap B^c, \qquad (A \cap B)^c = A^c \cup B^c.$$

Example in words

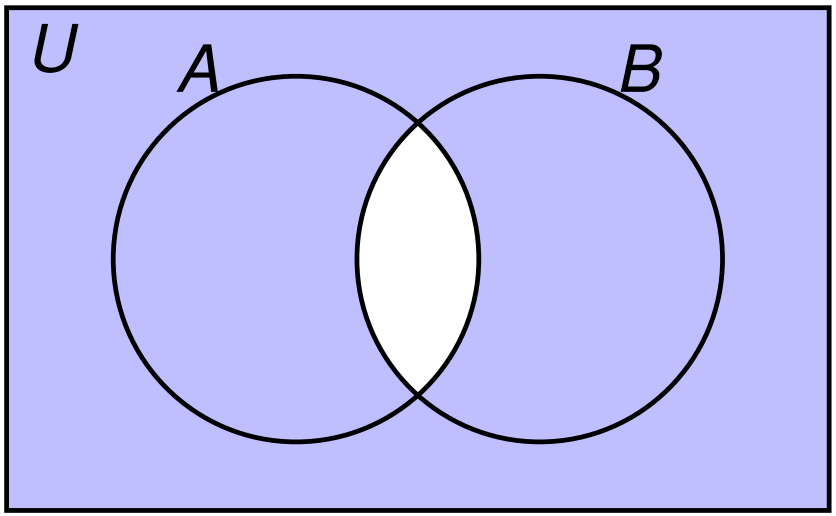
"Not (plays chess or plays Go) means "does not play chess and does not play Go.

Test both identities with $U = \{1, 2, 3, 4\}$, $A = \{1, 2\}$, and $B = \{2, 3\}$.

De Morgan's laws in pictures



$$(A \cup B)^c = A^c \cap B^c$$



$$(A \cap B)^c = A^c \cup B^c$$

Left: outside both circles.
Right: everywhere except the overlap.

Check: In the right picture, why is the part belonging only to A shaded?

De Morgan's laws — example

Here $A^c = \{3, 4\}$ and $B^c = \{1, 4\}$.

$$\begin{aligned}(A \cup B)^c &= \{1, 2, 3\}^c = \{4\},\\ A^c \cap B^c &= \{3, 4\} \cap \{1, 4\} = \{4\},\\ (A \cap B)^c &= \{2\}^c = \{1, 3, 4\},\\ A^c \cup B^c &= \{3, 4\} \cup \{1, 4\} = \{1, 3, 4\}.\end{aligned}$$

This verifies this example. A proof must work for every x and every choice of A, B.

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Cardinality

Definition for finite sets

The **cardinality** $|A|$ is the number of elements of A.

$$|\{2,4,6\}| = 3, \qquad |\varnothing| = 0, \qquad |\{\varnothing\}| = 1.$$

Counting overlap

For finite sets,

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

The subtraction corrects the elements counted in both sets.

How many elements?

Count the objects at the outermost level of braces.

Discuss with a partner

- $|\{1, 1, 2\}|$
- $|\{\varnothing, \{\varnothing\}\}|$
- $|\{\{1, 2, 3\}\}|$
- $|\{1, \{1\}, \{\{1\}\}\}|$

How many elements? — discussion

Count the objects at the outermost level of braces.

- 2: the elements are 1 and 2.
- 2: the empty set and a singleton are distinct.
- 1: the one element is the set $\{1, 2, 3\}$.
- 3: each successive pair of braces creates a different object.

The power set

Definition

The **power set** $\mathcal{P}(A)$ is the set of all subsets of A:

$$\mathcal{P}(A) = \{B : B \subseteq A\}.$$

If $A = \{a, b\}$, then

$$\mathcal{P}(A) = \{\varnothing, \{a\}, \{b\}, \{a, b\}\}.$$

Every element of $\mathcal{P}(A)$ is itself a set.

Counting all subsets

If A has n elements, then

$$|\mathcal{P}(A)| = 2^n.$$

Why?

To choose a subset, make one of two choices for each element of A: include it or leave it out.

- The empty set corresponds to leaving everything out.
- A itself corresponds to including everything.
- List the eight subsets of $\{a, b, c\}$.

Counting all subsets — list

$$\begin{aligned}\mathcal{P}(\{a,b,c\}) = \{&\varnothing,\\ &\{a\},\{b\},\{c\},\\ &\{a,b\},\{a,c\},\{b,c\},\\ &\{a,b,c\}\}.\end{aligned}$$

There are $1+3+3+1 = 8 = 2^3$ subsets.

Power-set check

Let $A = \{1, 2\}$.

Discuss with a partner

- $1 \in \mathcal{P}(A)$
- $\{1\} \in \mathcal{P}(A)$
- $\varnothing \in \mathcal{P}(A)$ and $A \in \mathcal{P}(A)$
- $\{\{1\}\} \subseteq \mathcal{P}(A)$

Power-set check — discussion

Let $A = \{1, 2\}$.

- False: 1 is not a subset of A under our convention.
- True: $\{1\} \subseteq A$.
- Both true: $\varnothing \subseteq A$ and $A \subseteq A$.
- True: its only element $\{1\}$ belongs to $\mathcal{P}(A)$.

The power set of the empty set

$$\mathcal{P}(\varnothing) = \{\varnothing\}.$$

Reason

The empty set has exactly one subset: itself.

$$\mathcal{P}(\{\varnothing\}) = \{\varnothing, \{\varnothing\}\}.$$

Check

What are the cardinalities of these two power sets?

They are 1 and 2.

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Ordered pairs

An **ordered pair** (a, b) records a first entry and a second entry.

$$(a, b) = (c, d) \quad \Longleftrightarrow \quad a = c \text{ and } b = d.$$

- $(1, 2) \neq (2, 1)$.
- $\{1, 2\} = \{2, 1\}$.
- $(1, 1)$ has two entries, while $\{1, 1\} = \{1\}$ has one element.

A pair of dice rolls records order. A set of observed values forgets it.

Cartesian products

Definition

$A \times B = \{(a, b) : a \in A \text{ and } b \in B\}$.

If $A = \{1, 2\}$ and $B = \{a, b\}$,

$$A \times B = \{(1, a), (1, b), (2, a), (2, b)\}.$$

- For finite sets, $|A \times B| = |A|\,|B|$.
- For two dice, the outcome set is $\{1, \ldots, 6\} \times \{1, \ldots, 6\}$.
- There are 36 ordered outcomes.

Product check

Let $A = \{1, 2\}$ and $B = \{3\}$.

Discuss with a partner

- $(1, 3) \in A \times B$
- $\{1, 3\} \in A \times B$
- $A \times B = B \times A$
- $A \times \varnothing = \varnothing$

Product check — discussion

Let $A = \{1, 2\}$ and $B = \{3\}$.

- True: $1 \in A$ and $3 \in B$.
- False: the elements of the product are ordered pairs, not these unordered sets.
- False: $(1, 3)$ belongs only to the left side.
- True: no second entry can come from the empty set.

Outline

1. SoPS Lecture 17
   - Objects and statements
   - Describing sets
   - Subsets and the empty set
   - Sets containing sets
2. Set algebra
   - Union, intersection, and difference
   - Counting and power sets
   - Ordered pairs and products
   - Quantifiers and proofs
3. Why is it called naive?
   - Russell's paradox

For every and there exists

$$\forall x \in A, \quad x \in B$$

means that every x in A belongs to B.

This is exactly $A \subseteq B$.

$$\exists x \in A \text{ such that } x \in B$$

means that at least one element belongs to both sets.

This is exactly $A \cap B \neq \varnothing$.

What is missing?

Which lines have a fixed truth value as written?

Discuss with a partner

- $x \in \{1, 2\}$
- $\forall x \in \{1, 2\},\ x < 3$
- $\exists x \in \{1, 2\},\ x > 3$
- $\{x \in \mathbb{Z} : x < 3\}$

What is missing? — discussion

Which lines have a fixed truth value as written?

- An open sentence: meaningful, but depends on x.
- A true statement: check both 1 and 2.
- A false statement: neither possible choice works.
- A set expression, not a statement. The x is bound inside the description.

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Can every description define a set?

So far, we have selected elements from an existing set:

$$\{x \in U : P(x)\}.$$

A tempting extra step

Can every condition define a set of all objects satisfying it, without an existing set to select from?

Consider the proposed set

$$R = \{S : S \text{ is a set and } S \notin S\}.$$

Russell's paradox

Suppose $R = \{S : S \text{ is a set and } S \notin S\}$ were a set.

Does $R \in R$?

- If $R \in R$, then R must satisfy the defining condition, so $R \notin R$.
- If $R \notin R$, then R satisfies the defining condition, so $R \in R$.

The conclusion

This unrestricted description cannot define a set. The problem is the assumption that every condition produces a set.

Why this mattered to Frege

- Gottlob Frege sought to derive arithmetic from logic.
- His *Grundgesetze der Arithmetik* appeared in two volumes, in 1893 and 1903.
- Its Basic Law V governed when two value-ranges are equal. Together with his logical system, it allowed the contradiction.
- The issue reached the foundations of his project, rather than an isolated calculation.

Source: Russell Archives, letter of 16 June 1902 and editorial note.

Source: Frege to Russell, 22 June 1902.

Timeline: the letter arrives

1901 Russell discovers the contradiction.

16 June 1902 He writes to Frege about it.

22 June 1902 Frege replies and acknowledges the problem.

1903 Frege publishes volume II with an appendix addressing the contradiction. Russell publishes *The Principles of Mathematics*.

Source: Russell's Paradox, historical discussion.

Source: Russell Archives, 53946.

Source: Original correspondence.

Russell's wonderfully understated letter

16 June 1902

"There is just one point where I have encountered a difficulty."

Russell praises Frege's work, then explains that either answer to the self-membership question gives its opposite.

His conclusion: being describable does not guarantee that a collection forms a set.

Check: What assumption does our proof force us to reject?

Source: Russell to Frege, Woodward translation, Russell Archives.

Frege's reply: 22 June 1902

An actual reaction, in translation

"Your discovery of the contradiction has surprised me beyond words"

Frege identifies Basic Law V as the source of the difficulty and worries about the foundations of arithmetic.

He also hopes the discovery may produce progress in logic and says he must think further about it.

Source: Frege Reader, p. 254, excerpt reproduced by Dirk Schlimm.

Source: Frege's original German letter.

Frege receives the letter



A totally real scene that definitely wasn't AI-generated
"Scheiße!" means "hello!"

The real manuscript was already going to press

- The second volume was nearing publication when the letter arrived.
- Frege added an appendix explaining that a foundation of the finished work had been shaken.
- He proposed a repair to Basic Law V. The repair did not successfully restore the intended foundation.
- The cartoon exaggerates the scene. The admission in print was real.

Source: Russell Archives, editorial note quoting the appendix.

Source: Frege's response.

Russell's later tribute

Russell later praised Frege's response as an extraordinary example of intellectual honesty.

He emphasized Frege's willingness to face a serious flaw in work that had taken years to develop.

Discussion

What would it look like to respond this way when somebody finds an error in your own solution?

Source: Russell's later recollection, quoted from van Heijenoort, p. 127.

Hilbert's reaction: already known in Göttingen

- On 7 November 1903, David Hilbert wrote to Frege after receiving volume II.
- He reported that the contradiction was already known in Göttingen, and credited Ernst Zermelo with discovering it independently several years earlier.
- For Hilbert, such problems called for clearer axioms and attention to consistency.

The discovery did not reach everyone at the same time or have the same impact on every foundational project.

Source: Volker Peckhaus, Pro and Contra Hilbert (2005), pp. 203–205.

How mathematicians responded

1908: Zermelo An axiomatic set theory restricts set formation.

1908: Russell A theory of types controls which objects can apply to which others.

1910–1913 Whitehead and Russell publish the three volumes of *Principia Mathematica*.

These were substantial rebuilding projects, developed over years.

Source: Zermelo's 1908 axiomatization.

Source: Russell's theory of types and Principia Mathematica.

Two ways to block the construction

Axiomatic set theory: restrict set formation

Select from an existing set U:

$$R_U = \{x \in U : x \notin x\}.$$

This set exists, but R_U cannot belong to U.

A simple type hierarchy: restrict expressions

A set at one level has elements at a lower level.
Self-membership $x \in x$ fails the type rules.

These approaches impose different rules for meaningful expressions.

Source: John D. Norton, Naive Set Theory, type and axiomatic approaches.

Why restricting to a universe helps

Let $R_U = \{x \in U : x \notin x\}$.

Suppose $R_U \in U$. Then its defining condition gives

$$R_U \in R_U \quad \Longleftrightarrow \quad R_U \notin R_U,$$

which is impossible. Hence $R_U \notin U$.

What follows?

Every proposed set U misses at least one set. There is no set of all sets in this axiomatic framework.

We can select within U without assuming U contains everything.

Paradox check: what is the conclusion?

- "All mathematics is false.
- "The proposed set R exists, but its membership is mysterious.
- "Unrestricted set formation leads to a contradiction.
- "Every discussion of self-reference is meaningless.

Which conclusion does our argument justify?

Paradox check — discussion

The argument justifies the third claim: unrestricted set formation leads to a contradiction.

It does not establish that all mathematics is false, or give us a mysterious existing set R.

Self-reference itself is not always meaningless. Its status depends on the language and rules we choose.

Connection to our first questions

We must distinguish a meaningful formula, an existence claim, and a construction our rules actually permit.

Historical sources

- Russell to Frege, 16 June 1902. Russell Archives, McMaster University, transcription and translation.
- Frege to Russell, 22 June 1902. Original German, Bibliotheca Augustana.
- Russell's Paradox. Stanford Encyclopedia of Philosophy, especially historical sections.
- V. Peckhaus, Pro and Contra Hilbert: Zermelo's Set Theories (2005), pp. 203–205.
- J. D. Norton, Naive Set Theory. Frege's appendix and foundational responses.

The meaning of naive set theory

- We develop the working language of sets through examples and ordinary reasoning.
- We do not assume that every imaginable collection is a set.
- Modern axiomatic set theory controls which set constructions are allowed.
- Selecting elements from an existing set is a safe construction: $\{x \in U : P(x)\}$.

Check

Why is $\{n \in \mathbb{Z} : n \notin n\}$ outside our classroom language?
We have not defined membership in an ordinary number.

Final thoughts

Reflection

- How do $x \in A$ and $\{x\} \subseteq A$ relate?
- Why is $\varnothing \subseteq A$ always true?
- Give a meaningful false statement and an undefined expression.
- Find $\mathcal{P}(\{\varnothing, 1\})$.

Submit your responses by **Tonight!**